FlashANNS: Hiding NAND from Graph Search on Flash-Backed CXL Memory

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Abstract

Billion-scale graph ANNS needs a flash-priced corpus and a hot path a query host can score from. Block-SSD systems such as DiskANN hide NAND behind asynchronous I/O and PQ navigation, but remain a per-host block serving unit. Full-in-memory CXL-DRAM systems such as CXL-ANNS and Cosmos remove the flash miss entirely and price the cold tail at DRAM. Flash-backed CXL memory is the third point: full-precision vectors remain on Flash, a 4 GiB CXL-DRAM page cache exposes memory-semantic access, and host DRAM keeps the adjacency graph, compact PQ codes, and a smaller scoring window. A unified VA does not make search usable. The runtime question is what must be hidden so that graph walk does not wait on NAND.

We present FlashANNS, a graph-ANNS runtime that splits that contract. *Score hide* permits a host-cache or CXL-DRAM-cache hit, but no scoring access may trigger or wait for a Flash fill. *Query hide*—the timed path also does not wait for a fill—is the oracle, not a result we claim. Host-side PQ navigation first commits a bounded rerank set; wise prefetching then fetches only its missing full-precision vector pages, using co-use-aware placement and dense page requests to increase useful bytes per read. Continuous batching overlaps one query's committed-page materialization with navigation or reranking from other queries, sustaining bounded Flash parallelism without changing per-query search semantics. The evaluation separates host-cache hits, CXL-DRAM hits, and Flash fills at iso-recall, and compares the three-tier path with demand access, block-SSD search, and a separate full-corpus CXL-DRAM Oracle. A same-machine PipeANN block path provides a named block-I/O comparison at matched recall. We evaluate a dual-NVMe Flash backing through the software memory-semantic manager `vmem_sw`; we do not equate that manager with a product CXL.mem SSD or claim 160 GB/s, and we treat multi-host one-copy as an unevaluated Shared-FAM deployment model.

CCS Concepts: • **Computer systems organization** → **Secondary storage organization**; • **Information systems** → **Nearest-neighbor search**; • **Software and its engineering** → *Memory management*.

Keywords: CXL, CXL-SSD, approximate nearest neighbor search, graph ANNS, memory-semantic flash, residency, prefetch, continuous batching

1 Introduction

Graph-based approximate nearest neighbor search (ANNS) is a production serving primitive: a query walks a proximity graph and scores neighbors until a beam budget is spent. At billions of vectors the full-precision embeddings no longer fit in host DRAM, even when the more compact graph and PQ metadata do, so the vector payload must live behind another tier. CXL makes that tier addressable with `load/store`. When the whole corpus fits in CXL-DRAM, prior systems already offer a usable service—CXL-ANNS and Cosmos [15, 20] never stall on flash. This paper is about the next device. Flash-backed CXL memory exposes Flash capacity through a bounded CXL-DRAM page cache on a CXL.mem-style address space [6, 22, 41, 42]. A query host may keep a still smaller software cache, but a miss in the CXL-DRAM cache costs tens of microseconds because it fills from Flash. To our knowledge, FlashANNS is the first *graph-ANNS serving runtime* that operates directly over an SSD-backed CXL address space and enforces *score hide*. The host computes from its small cache or from a CXL-DRAM-cache hit; no scoring access may trigger or wait for a Flash fill. We do not claim the first CXL-DRAM ANNS, the first CXL-SSD attached to a vector database, or a product CXL.mem SSD.

Two published lines each solve one side of the capacity–latency tension and tax the other (Figure 1). DiskANN [31, 32] and pipelined descendants such as PipeANN [13] are excellent *block-SSD* ANNS: PQ navigation, sector layout, and asynchronous I/O hide NAND behind a per-host cache. Their serving unit remains CPU + DRAM working set + a local SSD index. Scaling the service across hosts commonly replicates or shards the flash-resident index. They hide *block I/O*, not a memory-semantic hierarchy with two managed cache levels. CXL-ANNS and Cosmos put the graph and embeddings in a CXL-DRAM pool. Search never sees flash, but the cold tail $D - H$ is priced at DRAM [5, 19]. We build a third service because neither line is the right contract for flash-backed CXL memory: full-precision vectors should stay Flash-priced, compact search information should guide movement from host DRAM, and the runtime must hide Flash fills from the scoring instruction. Multi-host one-copy under Shared FAM is a deployment path we do not evaluate (§2).

Making that service real is hard because the device is not slightly slow DRAM and not a DiskANN SSD.

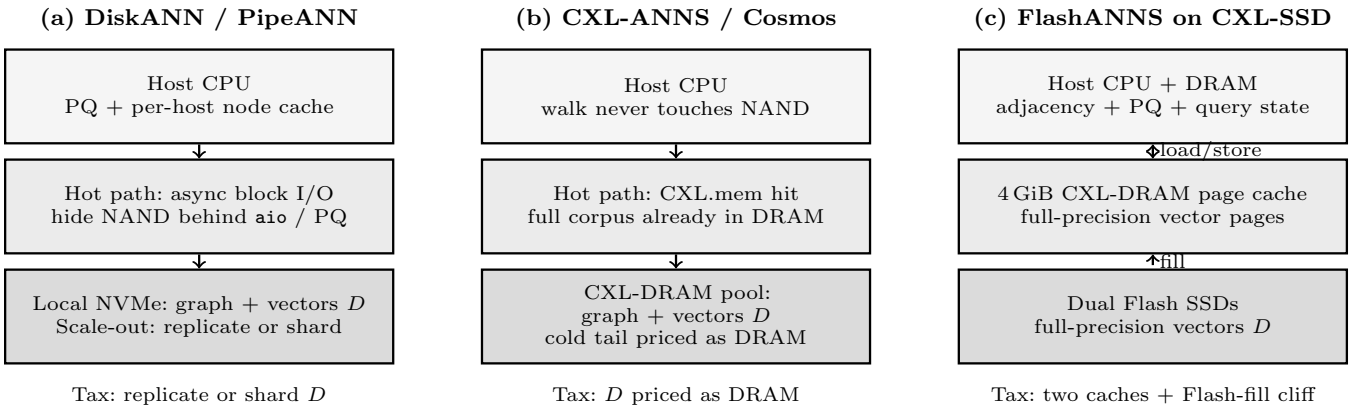

Figure 1. Two published deployments and the third point this paper serves. Flash-backed CXL memory is the *opportunity*: Flash-priced D, a 4 GiB CXL-DRAM page cache, and a small host cache. FlashANNS is the *runtime*: a wise prefetcher plus continuous batching so scoring does not wait on NAND. Host DRAM holds search information; the two lower tiers cache and store the full-precision payload. Shared FAM is not a measured link.

Three hardware–search interactions break conventional designs (§3). *(C1) The miss cliff.* A CXL-DRAM-cache hit is a DRAM-class access; on our measured path, a Flash fill takes about 89 μs, or 130–320× the latency of a CXL-DRAM item load. Graph walk multiplies the tax by hops × degree, so treating the composed hierarchy as uniform mmap'd memory makes search miss-bound. *(C2) Prefetch must be wise.* Neighbor IDs are not sequential. OS-style page admission and synchronous 2-hop prefetch bring records the walk will not use, pollute the 4 GiB cache, and can drop QPS by 3–5×. Wrong pages cost more than a missed page. *(C3) One query cannot fill the device.* After host-side PQ navigation commits a rerank set, one query cannot rerank until that set's missing vector pages are materialized. The query exposes one finite Flash wave and then waits, leaving device parallelism unused even when bandwidth remains. That dependency is local to the query: bounded cross-query overlap can cover its wait with another query's navigation, materialization, or reranking, but cannot make the single query as fast as an in-window oracle.

FlashANNS is a runtime for those three constraints, not a new graph index. A placement premise comes first: host DRAM holds graph adjacency and compact PQ codes, while full-precision vectors remain in the Flash-backed address space behind a bounded CXL-DRAM page cache. The first mechanism is a *wise prefetcher*. After PQ navigation commits the rerank candidates, it maps only their missing full-precision vectors to unique pages and issues those pages before exact scoring. An offline co-use-aware layout, online page deduplication, and density-bounded coalescing increase the fraction of each transferred page that reranking actually consumes; the prefetcher neither walks a speculative 2-hop neighborhood nor scores unused siblings to inflate utilization. The second mechanism is *continuous batching*. It overlaps one query's committed-page materialization with navigation or reranking from other queries, sustaining bounded Flash parallelism without changing per-query search semantics. Prefetch and batching may trigger Flash fills; the scoring thread may access the host cache or CXL-DRAM, but may not trigger or wait for Flash. We call that split *score hide*. The stronger contract—*query hide*, no wait for a fill on the timed path—is the oracle, not a result we claim. Our evaluation therefore reports host-cache hits, CXL-DRAM hits, score-triggered Flash fills, and critical-path Flash wait before reporting iso-recall throughput. The separate full-corpus CXL-DRAM Oracle removes Flash from the timed path; it is not a second active copy in the normal 4 GiB-cache configuration.

This paper makes four contributions:

1. **A third serving point, scoped.** We place graph ANNS on a three-tier host-cache/CXL-DRAM-cache/Flash hierarchy and separate the *opportunity* (Flash-priced D) from the *system* (hide Flash fills from scoring). We do not claim the first CXL-DRAM ANNS or an evaluated Shared-FAM deployment (§2).
2. **Why naive CXL-SSD ANNS fails.** We measure the miss cliff, unwise page/prefetch admission, and single-query queue collapse, and we turn those negatives into design constraints (§3).
3. **A wise prefetcher plus continuous batching.** The prefetcher waits for candidate commitment, fetches only missing rerank-vector pages, and shapes them for high useful-page occupancy; batching overlaps those precise materialization

waves with independent work from other queries under bounded concurrency. Score hide requires zero score-triggered Flash fills, while the full-corpus CXL-DRAM Oracle remains a separate upper bound (§4, §5).

4. **An iso-recall evaluation of hide, not just QPS.** We report the three-level hit/fill breakdown and critical-path Flash wait before iso-recall latency and throughput. Same-machine PipeANN is a named block-I/O comparison, not a hidden bar (§6).

2 Background

2.1 Graph ANNS walk

A graph ANNS index (Vamana [32], HNSW [25], or NSG [7]) stores, for each vector, a neighbor list of degree R. Search keeps a candidate set of width L (beam / `efSearch`). A *hop* expands the closest unread candidate u: the walk reads $N(u)$, scores those neighbors, and inserts the best unseen IDs back into the candidate set. The query returns the top-k after a hop or candidate budget is spent. Product quantization and its optimized variants compress the vectors used for approximate candidate ranking [8, 16]; FAISS and ScaNN provide widely used implementations of this quantized-search design space [14, 17].

One expand touches two kinds of bytes. Neighbor *IDs* are small and can live in host DRAM. Neighbor *vectors* are large: on Text2Image-10M each is 800 B, so scoring $N(u)$ moves about $R \times 800$ B if every neighbor is new. A typical hop is:

1. pick the closest unread candidate u from the beam;
2. read $N(u)$ (IDs only);
3. score the unseen neighbors' full-precision vectors;
4. insert the best new IDs into the candidate set.

Two facts follow for a memory hierarchy. First, the walk is *pointer-chasing*: step 3 of hop i decides which vector pages hop $i{+}1$ will need. Second, IDs need not live with vectors—the adjacency list can sit in host DRAM while embeddings sit on a slower tier. Recall depends on which useful neighbors the budget scores; stall rises with how many of those scores miss the fast tier. Widening L scores more nodes and cools the working set; that coupling is a measurement in §3, not a claim here.

2.2 The block-SSD serving stack

DiskANN and its maintained family [21, 31, 32] are the reference block-SSD design. Full-precision vectors and adjacency share 4 KB-aligned NVMe sectors. DRAM-resident PQ codes rank candidates without fetching full-precision vectors; an `aio` beam issues several sector reads at once; a static node cache retains entry and frequently visited vertices. PQ navigation exists to *avoid* full-precision reads, not to make NAND faster. Asynchronous I/O *hides* remaining reads by overlapping them with other work. Because graph and vectors share a sector, one 4 KB read is bound to whatever else the layout packed into that page.

Table 1. Latency tiers that a graph walk can see. CXL-DRAM is the published range; NAND fill is the order on the path we measure. The cliff itself is §3.

Access	Scale	Where
Host DRAM	$\sim 100\,ns$	graph / PQ metadata
Score-window hit	DRAM class	resident vectors
CXL-DRAM	140–410 ns	pooled graph / vectors
NAND fill	tens of μs	miss → flash

PipeANN [13] keeps that layout, overlaps compute with I/O inside one query, and can interleave queries to hold device queue depth. That cross-query interleaving is the block analogue of the continuous issue used later: its objects are I/O requests and sectors. The same idea will reappear on CXL-SSD as fetch-level issue (§4), against a faulting address space rather than `aio`.

2.3 CXL-DRAM ANNS

CXL.mem maps device HDM into the host physical address space [6]. Published Type-3 CXL-DRAM latencies sit in the 140–410 ns range [15, 33]. CXL-ANNS places the graph and embeddings in a CXL-DRAM pool and hides that far-memory delay with a relationship-aware cache and DSA-assisted prefetch. Cosmos [20] further offloads traversal and distance to device-side compute. Second-tier and residual designs [5, 28, 43] likewise assume a memory-class far tier without a per-miss NAND fill.

The interface is `load`/`store`; the capacity assumption is that D fits in CXL-DRAM. A service on that line is already usable in the sense that search never waits on flash. Prefetch and cache policies tuned for hundreds of nanoseconds do not transfer to a NAND fill of tens of microseconds. We keep the detailed contrast in §7.

2.4 Flash-backed CXL memory

In the device model studied here, a CXL-SSD pairs NAND capacity with a byte-addressable window [18, 22, 39, 41, 42]. Figure 1c is the deployment we serve. A *score-window hit* is served at DRAM-class latency. A *miss* is a NAND fill into that window, then a retry. The programming model looks like memory; the miss tax looks like flash. Table 1 places the three published tiers next to the NAND fill this paper must hide. The measured cliff belongs in §3; the table only fixes the orders of magnitude.

The window is bounded, and the admission unit is a page. On Text2Image-10M a vector is 800 B, so a 4 KB page holds several records—or several unused siblings. A blocking `load`/`store` miss and an explicit prefetch ioctl are both fills; they differ in *who waits*. If the scoring thread issues the fill, NAND is on the critical path. An early prefetch can overlap the fill; only a fill that completes before scoring leaves the scoring path. That split is the runtime contract in §4, not a result.

The prototype we measure is a software memory-semantic node (`vmem_sw`) in front of NVMe, with the score window implemented as host DRAM (`mbind`). It preserves fill semantics (a miss moves NAND bytes into the window), but not product CXL.mem timing. Hardware and generic OS work on this tier does not manage graph-ANNS residency; those systems belong in §7. Implementation limits belong in §5. Direct-attach CXL-SSD already gives one host flash-priced memory semantics; that is the deployment we measure. Multi-host one-copy under CXL 3.x Shared FAM is a premise we do not evaluate.

3 Motivation

Section 2 defined the walk and the fill. This section measures what happens if that fill is treated as a slightly slow DRAM load or as a DiskANN sector read. Each measurement asks two questions: does the effect exist, and what throughput or NAND-traffic penalty does it cause? The three results are the constraints Intro named C1–C3 (Table 2).

3.1 The miss cliff

A host `load` of one vector from CXL-DRAM is a few hundred nanoseconds. A matched-size CXL-SSD window-miss read is a NAND fill (Figure 2). On Text2Image-10M (800 B) the p_{50} is 0.28 μs versus 89 μs (∼320×). On a 10M-item LAION slice (2048 B) it is 0.67 μs versus 89 μs (∼130×). The miss is a 4 KB fill, so record size barely moves it; the CXL-DRAM hit scales with bytes.

A graph walk can pay that tax across hops and newly scored neighbors. Once a modest fraction of scores miss, wall-clock time is in the NAND path, not in distance arithmetic. DiskANN already knew the cliff for block I/O. The relevant distinction is that a memory-semantic interface does not remove it: `load`/`store` only changes *how* a miss is issued (§3.3). Treating CXL-SSD as memory-mapped DRAM therefore makes search miss-bound. The runtime contract that follows is not "make NAND faster"; it is that the scoring instruction must not wait on a fill.

3.2 Prefetch must be wise

Neighbor IDs are not sequential in the address space, so the next vector page is not the next byte. Coalescing

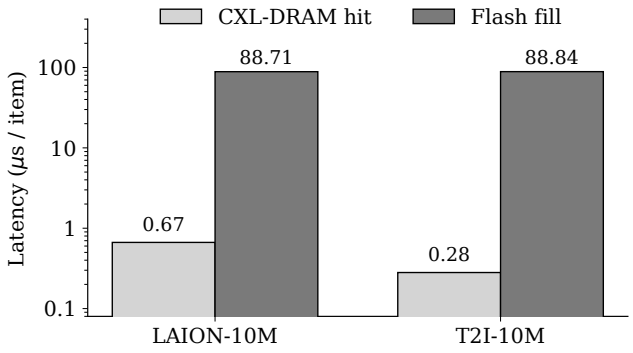

Figure 2. One-item host load: CXL-DRAM hit versus CXL-SSD cache miss. CXL-DRAM is Montage `/dev/dax0.0` (`clflush` then load). CXL-SSD is a `vmem_sw` window miss. T2I-10M uses the staged 800 B vectors. LAION-10M is the first 10M items of the 25M dump (d=512); the miss bar matches that item size on the same SSD path.

four 4 KB pages into one 16 KB admission unit therefore brings siblings the walk will not score. On a high-dimensional corpus where a record already fills most of a page, per-record admission versus page admission splits QPS by ∼3.6× and NAND reads by ∼3.7× (Figure 3a). At small dimensions several records share a page and the gap narrows; page-cache thinking is still the wrong default here.

Synchronous two-hop prefetch rooted at the top-1, top-2, or top-8 frontier candidates does not fix the cliff. Hit rate is almost unchanged, NAND reads multiply, and QPS drops by ∼4–5× at the widest setting (Figure 3b). Precision (*used*/*promoted*) falls as coverage grows; left-over pages evict useful ones from a small window. Wrong pages cost more than a missed page. A page can be necessary yet still waste most of its vector slots. The useful policy therefore delays full-precision movement until the approximate search commits its rerank set, then fetches only that payload and packs co-used vectors together—not "prefetch harder."

Widening L can become another form of unwise coverage. Under the fixed hop budget, recall and NAND reads both plateau from L=32 to 64. Extra width buys no measured quality benefit in that interval; quality and I/O cost must therefore be reported together.

3.3 One query cannot fill the device

Best-first search is dependency ordered even when its graph and PQ codes are host resident: approximate navigation must commit C_L before the runtime knows the exact full-precision payload to fetch, and reranking must wait until that payload is resident. A single query therefore exposes only one finite fill wave and then waits,

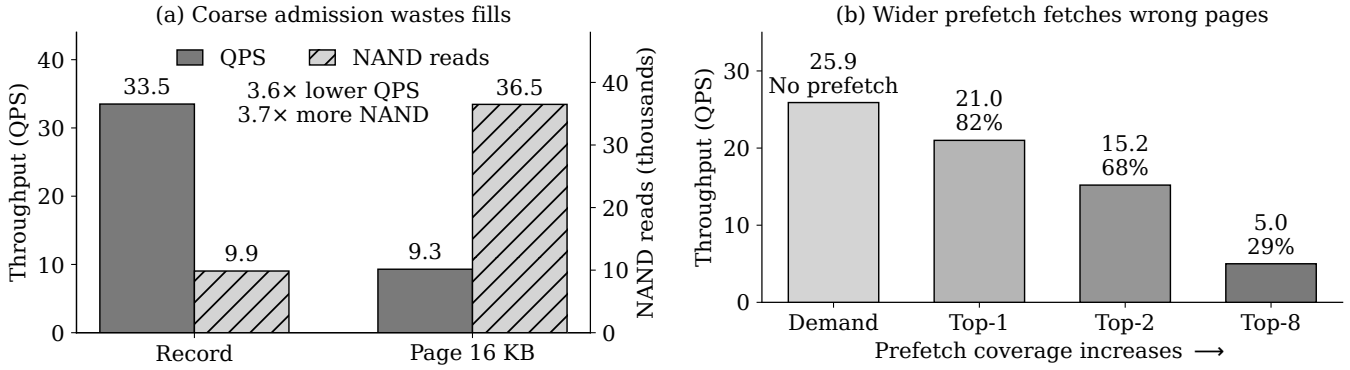

Figure 3. Unwise admission amplifies the cliff (LAION-200k, d=1024, hops=32). (a) A 16-KB admission unit generates 3.7× more NAND reads and cuts throughput by 3.6× relative to record admission. (b) Widening synchronous prefetch lowers promotion precision from 82% to 29% and throughput from 21.0 to 5.0 QPS; the unchanged 58% hit rate is omitted.

leaving device-queue bubbles before reranking (Figure 4). A block interface exposes I/O requests that a scheduler can interleave across independent queries. On CXL-SSD the programming model is a faulting address space. A blocking `load/store` (or a blocking `memcpy` from the mapping) is a synchronous NAND fault: effective queue depth collapses to ≈ 1 per thread. If a query waits for those fills before scoring, NAND latency remains on its critical path.

The idle device is a service-pipeline problem, not a bandwidth wall. The dependency is local to a query: while one query waits for its committed pages, another can navigate, materialize, or rerank. Bounded cross-query overlap can therefore sustain useful Flash work without changing which candidates any query visits or scores; the mechanism belongs in §4.

3.4 From measurements to constraints

Table 2 is the contract for §4. We do not treat CXL-SSD as uniform memory (C1): scoring must not touch NAND. Prefetch must be wise (C2): trigger at candidate commitment, fetch only the committed full-precision payload, and place co-used vectors so each page carries more data that reranking consumes. One query cannot fill the device (C3): bounded cross-query overlap must cover its materialization barrier with independent work from other queries.[1] Placement of neighbor IDs in host DRAM is a premise for C2, not a separate challenge: without it the prefetcher misses once just to learn whom to fetch.

[1]A small shared pin of entry vertices helps a narrow beam more than deepening one walk. That is an admission hint, not a positive LFU-hot-set module, and not a fourth challenge.

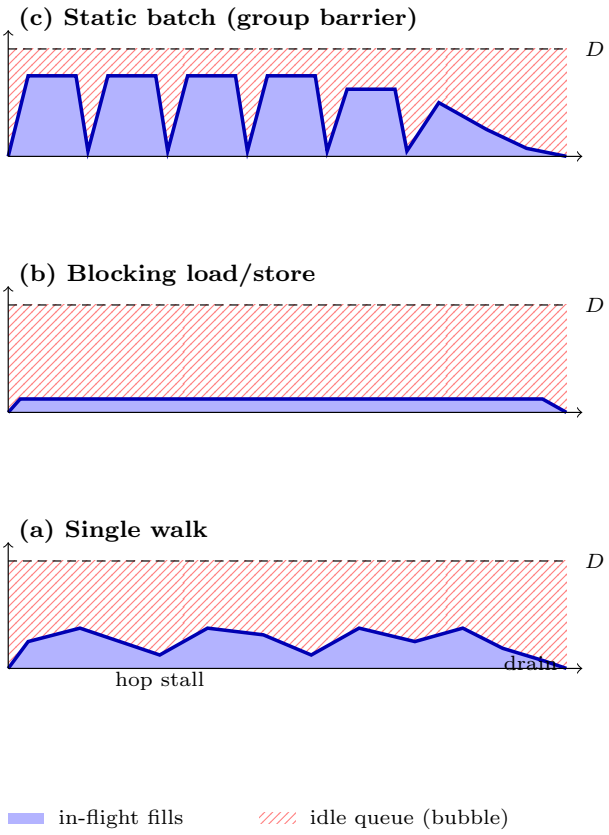

Figure 4. Pipeline bubbles on a faulting CXL address space (schematic). y is in-flight fills; D is the depth NAND parallelism needs. A single query cannot hold D; a blocking load/store is stuck at QD≈ 1; a static batch drains at the barrier. How the runtime issues across queries is §4.

Table 2. Measured pathologies become the three constraints Intro named. C1 uses the 10M-item probes in Figure 2; C2 uses LAION-200k; C3 follows from the dependency and issue path illustrated above.

ID	Exists + hurts	Constraint	FlashANNS answers
C1	Miss cliff ∼130–320×	Score only resident bytes	Score-hide contract
C2	Page ∼3.6×; blind prefetch ∼4–5×	Commit, select, and pack useful pages	Wise prefetcher
C3	One committed wave; blocking fill ⇒ idle Flash	Overlap independent queries	Continuous batching

4 Design

FlashANNS is a userspace ANNS runtime plus a thin residency plane in front of host DRAM, a bounded window that plays the role of CXL-DRAM, and a flash-backed CXL address space that holds D (Figure 5). Search is still best-first graph walk. The service contract is a *split*: prefetch and continuous batching may touch CXL-SSD; the scoring thread may not. A hop that computes a full-precision distance from bytes that were not already in the window is a hide failure, not a successful promote.

The wise prefetcher turns a committed PQ rerank set into one precise page batch before full-precision scoring. That precision does not remove the query's materialization barrier: its rerank still waits for its own batch. Continuous batching exploits the independence across queries to cover that wait with other useful work. The modules below separate these two roles and the evaluation tests them independently.

4.1 D1: Semantic placement

Table 3 is the default map. The graph adjacency and the PQ codebook are touched on every hop and are small relative to the full-precision corpus; they live in host DRAM so they never compete with vectors for the CXL-SSD window (F2, F7). Query state (candidates, visited set, thread-local arenas) is likewise host-local. Full-precision vectors default to the CXL-SSD address space. A bounded *hub / shared hot set* of vector pages is explicitly cached into the window or into CXL-DRAM (§4.3). Only full-precision vectors in a committed rerank set become prefetch candidates; a discovered frontier does not (F3).

This is a deliberate difference from DiskANN, where adjacency and vectors share SSD sectors. We pay a modest host-DRAM tax for the graph in order to stop the window from turning into an accidental adjacency cache.

Table 3. Default placement. **Fill-in:** bytes on the corpus you report (e.g., 25M × dim).

Object	Default tier	Why
Graph (R neighbors)	host DRAM	every hop; keep off the window
PQ / codebook	host DRAM	tiny, extremely hot
Query / visited	host DRAM	per-worker state
Full-precision vec.	CXL-SSD	capacity
Entry / soft-pin	window (tiny)	F7; LFU hot set demoted
Committed rerank vec.	window via prefetch	F3, F5; score only if resident

4.2 D2: Wise prefetch at candidate commitment

When: after commitment. The host first completes its approximate beam using the adjacency graph and PQ codes, without reading full-precision vectors from Flash. When that beam commits the bounded rerank set C_L, the prefetcher issues immediately: after uncertainty has been pruned, but before exact scoring dereferences any vector in C_L. This trigger deliberately gives up speculative lead time to avoid moving payload for candidates that will not survive. The single-query path may still wait for materialization to complete; the invariant is that the subsequent scoring loads do not themselves fault on Flash.

What: the committed payload. The residency plane maps the IDs in C_L to their full-precision vector pages, removes pages that are resident or already in flight, and deduplicates the rest. Every requested page is therefore justified by at least one vector that exact reranking will consume. Adjacency, PQ codes, speculative neighbors, and unused frontier records are not part of this wave. Because Flash moves 4 KiB pages, however, a justified page may still carry vector slots that this query will not score.

How: make each page useful. Offline, a co-use-aware layout groups vectors that disjoint training searches tend to request together and places the resulting pages into short runs. Online, exact page filtering avoids duplicate movement, and the runtime fills gaps only in a short, sufficiently dense run; it never expands a sparse request into a whole SSD stripe. This combination targets useful payload per read rather than batch size alone. Unlike generic remote-memory or device-side sequential prefetchers [23, 26], the request is justified by the committed rerank set rather than an inferred address stream. We distinguish page precision (a fetched page contains any scored vector), vector-slot utilization (scored slots

Fig. 6 placeholder — architecture:
top: Placement | Window admit | Prefetch+`cont` | L ctrl;
middle: residency plane (fill off crit path / score from window);
bottom: host DRAM (graph/PQ) | window = virtual CXL-DRAM | CXL-SSD.

Figure 5. FlashANNS architecture. From the host, scored bytes look like CXL-DRAM; capacity is CXL-SSD. The residency plane is the only path that moves full-precision bytes. Search never page-faults a neighbor "to be helpful." **Fill-in:** use shipped names (`PREFETCH_BATCH`, DramWindow); mark `vmem_sw`.

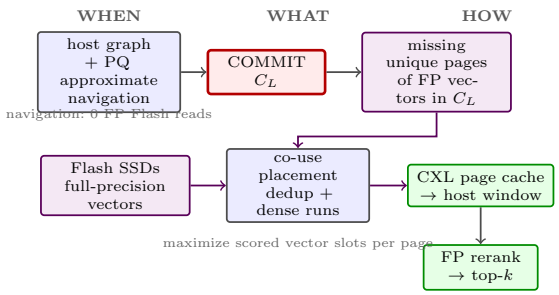

Figure 6. Wise prefetch answers three questions. *When*: after PQ navigation commits C_L. *What*: only missing unique pages of its full-precision vectors. *How*: co-use-aware placement, exact filtering, and dense short runs increase useful vector slots per Flash read.

over all slots read), and extent utilization (requested pages over pages transferred after coalescing); vector-slot utilization is the direct measure of real in-page use.

4.3 D3: Entry pin and soft-pin (hot set demoted)

Single-query deepening is a bad way to spend a small window (F4, F7). A 64 MiB shared LFU set raised hit rate and *destroyed* QPS on a warm protocol (14.3→2.4); we do not ship it as D3. What remains is a tiny hard pin of the entry subgraph (≤32 MiB) and a hop-decaying soft-pin of recently installed pages. Pinning is never implicit in a page fault. Cross-query reuse, if it happens, is because committed materialization left useful pages in the window—not because an LFU daemon is competing with search.

4.4 D4: Hide engine — wise prefetch and continuous batching

A query-local barrier. D2 creates one precise materialization wave after PQ navigation commits C_L, but the same query cannot rerank until that wave completes. Better page selection alone therefore cannot keep Flash busy.

Cross-query overlap. The barrier is local to a query, not to the service: while one query waits, another can navigate, materialize its own committed wave, or rerank resident bytes. Continuous batching interleaves these stages so host work and Flash access coexist (Figure 7). Like continuous scheduling in other staged serving systems [40], it admits work at a semantic boundary—candidate commitment rather than a model iteration.

Bounded admission. The runtime bounds in-flight query stages and admits another committed wave as capacity becomes available, avoiding both device bubbles and an unbounded work queue. It neither predicts future candidates nor implements adaptive queue-depth control.

Semantic invariance. Each query retains the same graph, PQ codes, beam budget, and visited state, and reranks only after its own pages are resident. Batching changes timing, but not its candidate set or recall.

4.5 D5: Observability

Every experiment in §6 is required to report hide counters first: fraction of full-precision scores whose bytes were already in the window (`score_from_window`), critical-path NAND wait (`crit_wait_ns`), and device-fill/wall overlap, then QPS at the iso-recall floor. Prefetch efficiency is reported as page precision, vector-slot utilization, and extent utilization; a page touched once is not evidence that all of its payload was useful. QPS without those counters is not a hide result.

What we refuse to hide. If a module cannot move `crit_wait_ns` or `score_from_window` in an ablation, it is removed from the contribution list rather than kept as decoration. An LFU hot set that raises hit rate and drops QPS is already in that bin.

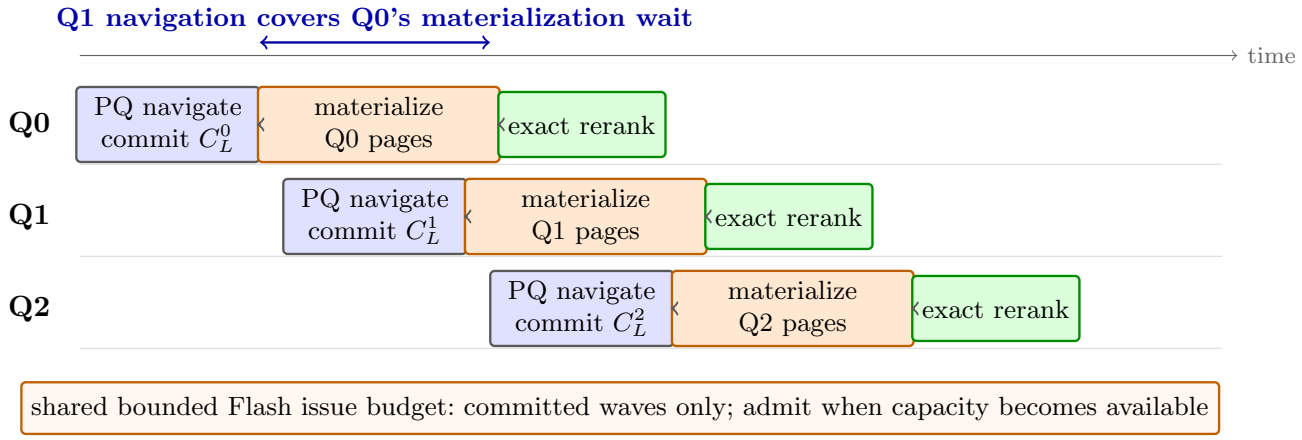

Figure 7. Continuous batching fills the service-level bubbles left by a single precise materialization wave. When Q0 waits for its committed pages, Q1 and Q2 contribute navigation, Flash materialization, or reranking under a bounded in-flight budget. Each query still crosses its own materialization barrier before reranking, so scheduling changes timing but not search semantics.

5 Implementation

FlashANNS is a userspace beam search with the adjacency graph and 64-byte PQ codes in host DRAM and full-precision vectors mapped from `/dev/vmem0`. The host completes PQ navigation, commits the L rerank candidates, removes resident and duplicate 4 KiB pages, and materializes the remaining pages into DramWindow before full-precision MIPS reranking. The explicit residency barrier leaves materialization wait on the query path, but prevents the subsequent scoring loads from triggering Flash faults. For continuous batching, workers maintain a bounded number of in-flight query stages and share asynchronous copy resources; completion-driven admission lets navigation or reranking from one query overlap another query's materialization without allowing either query to cross its own residency barrier early.

Measured prototype (2026-08-30, gpu01). We evaluate a userspace `vmem_sw` proxy over two CD8P SSDs: `/dev/vmem0` exposes a 28 GiB logical map backed by a 4 GiB device cache with 2 MiB striping. CXL-DRAM DAX and hardware fills into the FPGA BAR are absent, so DramWindow is host memory bound to NUMA node 1. The results therefore include proxy copying and cache-lock overhead and are not measurements of a shipping CXL.mem SSD.

Implemented scope. The prototype includes demand access, committed-candidate materialization, host graph and PQ-64 navigation, batched prefetch, co-use/extent layout, the iso-recall L picker, bounded cross-query scheduling, and a pinned entry subgraph of at most 32 MiB. A shared LFU hot set reduced warm-run QPS and is excluded.

Not claimed. Product CXL.mem SSD; HPS fill of the FPGA BAR; Shared FAM; a 160 GB/s DRAM–SSD engine; closed-loop F9 miss-chop; speculative candidate execution; or an adaptive queue-depth controller. DiskANN PQ-off is never placed on the PQ-on line.

6 Evaluation

We answer five questions:

- **Q1** Does naive CXL-SSD graph ANNS fail in the ways §3 claimed?
- **Q2** At iso-recall, does FlashANNS move FP scoring off the NAND critical path (hide), and what QPS remains vs. demand and vs. an in-window oracle?
- **Q3** Which Wise Prefetcher decisions are necessary, and does continuous batching raise useful device occupancy and throughput without changing recall?
- **Q4** Does hide appear only after warmup (service hide), or also on a true-cold reload?
- **Q5** What remains vs. an in-window oracle and vs. PipeANN, and is the residual honest?

6.1 Setup

We follow established ANNS evaluation practice [2, 30]: every performance comparison reports recall and uses matched-recall operating points when claiming a latency or throughput advantage.

Table 4 is the locked configuration. The primary metric is *iso-recall QPS* (and p99) at a stated recall@k floor. We always report NAND reads, promote bytes, and the three-level hit breakdown. A run that does not declare cold vs. warm is discarded: the window is small enough that accidental warm-up invents speedups.

Table 4. Locked evaluation platform (2026-08-30, `gpu01`).

Item	Configuration
Host CPU / DRAM	dual-socket Xeon, ∼64+64 GiB host DRAM
CXL-DRAM window	**absent**; 1 GiB DramWindow = `mbind` node 1
CXL-SSD	FPGA `nvmex`+CD8P; `/dev/vmem0`=`vmem_sw`
Window / cache	host scoring window; bounded `vmem_sw` cache (reported per run); T=8 uses eight 128 MiB proxy windows and a 100 MiB device cache
Block-SSD baseline	PipeANN on `/mnt/disk0` (`nvme0n1`), PQ on
Primary corpus	Text2Image-10M, 200-d, MIPS
Sensitivity	default vs. co-use/extent layout; L picker
Index	Vamana R=32, host graph + PQ-64 navigation, committed FP rerank
Target quality	recall@10 $\geq$ 0.92 (picker chose L=400)
Protocol	declared per row; frozen T=8 uses seed 42, nq=500, service-state cache; true-cold is isolated in Q4

Baselines. (1) **Demand:** CXL-SSD load/store, no promote. (2) **Page:** 16 KB admission. (3) **Sync-2hop:** the F3 policy. (4) **Static-wide:** large beam, no controller. (5) **Miss-chop:** cut beam on instantaneous miss (the F9 policy). (6) **DiskANN/PipeANN:** official block path, same machine, same corpus, same recall floor; PQ on unless a figure is labeled `--no_pq_nav`. (7) **Oracle-in-window:** pin the measured working set in the 1 GiB window, then search with no SSD fills (the hide upper bound). This is not DiskANN.

We never present a home-grown `O_DIRECT` reader as "DiskANN."

6.2 Q1: Naive CXL-SSD fails

Figure 2–4 are reproduced on the evaluation machine under the Table 4 protocol so that Design and Evaluation share one dataset. We expect the same qualitative shape: a two-order cliff, page admission and blind prefetch losing to demand, beam cooling the set, and blocking load/store stuck near QD≈ 1. If a finding does not reproduce, it is dropped from the intro, not explained away in a footnote.

6.3 Q2: End-to-end iso-recall (the money plot)

Figure 8 and Table 5 are the submission's main result. The frozen T=8 path sustains 689.32 QPS at 0.920 recall and 64.3% NAND occupancy (mean/P50/P99: 20.053/19.274/39.575 ms), with a 650–690 QPS repeat band; P95 was not logged and remains to be measured. Its throughput is 72.4% of the 952.07-QPS local-DRAM reference, but different nq and recall make that a diagnostic bound, not a matched speedup. The local reference is not the paper's full-corpus CXL-DRAM Oracle.

6.4 Q3: Each module is necessary

Figure 9 and Table 6 are the falsifiability test for contribution 3. The first two comparisons answer when and what to prefetch; the layout and coalescing comparisons answer how much of each fetched page is truly consumed; the final row must isolate continuous batching without changing the prefetch set, concurrency, or query protocol. The measured full-system row characterizes the frozen path, but it does not replace that matched causal ablation.

6.5 Q4: Cold hide versus service hide

Figure 10 is the service question. A production ANNS process is allowed a discarded warmup; a paper row is not allowed to confuse that with a true-cold reload. If `crit_wait` collapses only on the warm pass, we say so in the caption and do not migrate the sentence into §1. The one-copy capacity sketch of §2 remains a model; we do not plot it as a measured Q4.

6.6 Q5: Losses, overhead, and fairness

We report the cases FlashANNS loses or looks worse: warm working sets that already fit the window (async prefetch ≈ demand); low dimension where page≈record; the residual gap to PQ-navigated DiskANN; and software-proxy artifacts (global cache lock, QD ceiling). Host-DRAM consumed by the graph and PQ is a cost, not a footnote. CPU is reported because DiskANN on this class of machine is I/O-bound, not CPU-bound; if FlashANNS becomes CPU-bound we say so.

Takeaway. Q2–Q4 must be answerable from hide counters, not from QPS prose. If the money plot needs a verbal caveat that changes the claim, the claim in §1 is the thing to change.

7 Related Work

Table 7 is the comparison we want a reviewer to use. Prose below only records the one-line difference; it does not repeat the background.

Block-SSD ANNS. DiskANN, FreshDiskANN, and PipeANN optimize PQ navigation and I/O overlap on block devices [13, 31, 32]; SPANN and Starling explore hybrid or disk-resident layouts [4, 36]. SPFresh, Filtered-DiskANN, and Milvus extend this space to updates, filters, and serving [9, 35, 38]. Unlike an I/O scheduler, FlashANNS manages a memory-semantic window and

Fig. 9 money plot (hide first):
stacked `crit_wait` vs. compute at iso-recall;
twin axis QPS. Lines: Demand, committed-page baseline, full FlashANNS,
−prefetch, oracle-in-window, PipeANN T=1.

Figure 8. Hide at the primary recall floor. The money plot is critical-path NAND wait, not QPS alone. FlashANNS passes only if `crit_wait` approaches the in-window oracle. QPS vs. PipeANN is a named memory-semantic versus block-I/O comparison. Until hide counters pass we write "gap to demand," not "search cannot see flash."

Table 5. T2I-10M, L=400, seed 42, PQ-64 navigation, and committed FP rerank. Unmatched rows are diagnostic; NAND/promote values remain — until remeasured.

System	Recall@10	QPS	p99 (μs)	NAND reads/q	Promote B/q	Host DRAM
Demand CXL-SSD (P0)	0.944	1.64	1.3×10^6	—	—	1 GiB win.
Page admission	—	—	—	—	—	—
Sync 2-hop	—	—	—	—	—	—
Static wide beam	—	—	—	—	—	—
Miss-chop beam	—	—	—	—	—	—
DiskANN (PQ on)	—	—	—	—	—	—
PipeANN pipe T=1	0.925	57.6	1.7×10^4	—	—	PQ nav
Oracle host DRAM (nq=20)	0.950	66.5	4.8×10^4	0	—	9.6 GiB pop.
Oracle 1 GiB window (nq=20)	0.925	85.8	1.5×10^4	0	—	WS pinned
FlashANNS committed, default	—	—	—	—	—	—
FlashANNS wise, T=1 (nq=100)	0.923	116.29	1.344×10^4	—	—	shared win.
FlashANNS + CB, T=8 (nq=500)	0.920	689.32	3.958×10^4	—	—	8×128 MiB
Local Oracle, T=8 (nq=100)	0.923	952.07	2.872×10^4	0	—	full corpus

Every — cell is a placeholder. Replace with the measured value under the protocol in §6.1.

Fig. 10 placeholder — two panels:
(a) when/what: frontier issue vs. commit-time issue and broad vs. committed-page fetch;
(b) how: default vs. co-use layout, with/without dense coalescing; keep continuous batching as a separate bar.

Figure 9. Wise Prefetcher and continuous-batching ablation. Prefetch variants hold candidate IDs and recall fixed; the utilization panel distinguishes page precision, vector-slot utilization, and extent utilization. **Fill-in:** same recall floor as Figure 8.

the committed vector pages required before exact reranking.

CXL-DRAM and near-memory ANNS. CXL-ANNS, Cosmos, and second-tier/residual designs assume a DRAM- or NVM-priced far tier [5, 15, 20, 28, 43]; CXL-aware vector-database restructuring makes the same memory-class assumption [19]. Their ∼100 ns policies do not address tens-of-microseconds NAND behind a polluted window (F3), and FlashANNS does not require near-memory compute.

Far-memory runtimes and CXL tiering. Remote paging and application-integrated runtimes manage far capacity through paging, isolation, and prefetch [3, 12, 26, 29, 34]; CXL tiering places hot pages across DRAM-class tiers [1, 24, 27, 37, 44, 45]. Those policies react to generic locality; FlashANNS uses PQ commitment to identify reranking's full-precision payload.

Table 6. Wise Prefetcher decisions at the primary recall floor.

Variant	QPS	Slot	Extent	NAND occ.
Wise + continuous	689.32	55.91%	—	64.3%
Frontier issue	—	—	—	—
Broad payload	—	—	—	—
Default layout	—	—	—	—
No coalescing	—	—	100%	—
Wise only (T=8)	—	—	—	—

Every — cell is a placeholder. Replace with the measured value under the protocol in §6.1.

Fig. 11 — cold reload vs. 1 discarded warmup: y: `crit_wait`, score-from-window, QPS. If only warm hides, label it service hide.

Figure 10. True-cold (`vmem_sw` `cache_used` = 0) versus a discarded warmup then 100 queries. Hide that appears only after warmup is a service property, not a cold-start property. Shared-FAM provisioning stays a discussion model, not this figure.

Table 7. Closest systems and their defining distinction.

System	Slow tier	Primary distinction
DiskANN / PipeANN	block SSD	PQ navigation + async block I/O
Second-Tier / FaTRQ	far memory	coarse/fine far reads
CXL-ANNS / Cosmos	CXL DRAM / PNM	memory latency or compute offload
CXL-SSD hardware	NAND + window	generic memory-semantic access
FlashANNS	**CXL-SSD**	**committed pages + cross-query overlap**

CXL-SSD hardware and generic software. DirectCXL and memory pooling establish CXL expansion [10, 11]; CXL-enabled SSD, SkyByte, From-Block-to-Byte, and CMM-H study flash-backed memory [22, 39, 41, 42]. They do not jointly manage graph-ANNS residency, committed-vector materialization, and cross-query overlap.

8 Discussion and Limitations

Proxy versus product. The measured path is FPGA `nvmex` plus `vmem_sw`, not a shipping CXL.mem SSD; the FPGA BAR is not the evaluated window. Shared FAM and one-copy provisioning are likewise unmeasured; device IOPS would remain shared.

Hide is the service contract, not a fallback. A usable service means exact scoring reads resident bytes (`score_from_window` = 100%) and `crit_wait_ns` is DRAM-class. Graph/PQ live in host DRAM; after candidate commitment, wise prefetch materializes the rerank pages and continuous issue schedules those batches across queries. A window miss while scoring is a hide failure we close, not a reason to shrink the claim.

Block-path comparison. PQ navigation plus deep `aio` avoids most full-precision reads. FlashANNS also uses PQ navigation, but materializes a committed FP rerank set through a load/store hierarchy rather than issuing traversal records through block I/O. At T=8, FlashANNS reaches 689.32 QPS versus a 952.07-QPS same-machine local-DRAM reference; the 72.4% ratio is diagnostic because query counts and recall differ. The remaining gap includes Flash materialization and proxy copying under a cache lock; scaling beyond the fixed budget and removing that copy are engineering questions, not scheduling claims.

What would change our mind. If HPS makes the FPGA BAR a coherent miss window, we re-measure hide there and keep the same counters. If hide counters never leave NAND-class wait, contribution 3 shrinks to "overlap vs. demand" and the intro must not say the host is unaware of CXL-SSD.

9 Conclusion

Flash-backed CXL memory offers flash-priced capacity through a memory-semantic window, but is unusable if treated as slow DRAM or mmap'd DiskANN. FlashANNS separates search information from payload movement: wise prefetch materializes the missing pages of a committed rerank set, and continuous batching overlaps each query's materialization with independent work. Thus exact scoring reads resident bytes while the remaining query-local barrier stays explicit.

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